

Affective Mirage? Diagnosing Visual Grounding in Emotion Recognition with Vision Language Models

Supplementary Material

Anonymous WACV Datasets Track submission

Paper ID 2646

Abstract

Vision-language models (VLMs) have shown remarkable visual understanding capabilities, yet recent studies reveal a prevalent *MIRAGE* phenomenon: models may produce plausible predictions without faithfully relying on the underlying visual evidence. We investigate whether such behavior also exists in the visual emotion recognition task, which is inherently subjective and often relies on subtle and ambiguous visual cues, potentially making the task particularly susceptible to *MIRAGE*. Thus, we thoroughly examine open-source, closed-source, and domain-specific VLMs through a novel three-level diagnosis framework, named *Affective MIRAGE*. First, **general mirage** removes visual evidence to examine model behavior without reliable visual input. Second, **causal mirage** uses controlled intervention to examine how different visual evidence causally influences model predictions, together with mechanism-level analysis of visual grounding. Third, **certainty mirage** examines whether model uncertainty follows human disagreement on ambiguous emotions. Our findings reveal three characteristic behaviors of current VLMs in emotion recognition: 1) **without visual evidence, all evaluated VLMs maintain a nearly 100% mirage rate**, while prediction collapse exists toward label priors, 2) the causal dominance of subject evidence is over scene context under counterfactual interventions, and 3) uncertainty alignment with human affective ambiguity is insufficient. To our knowledge, this is the first systematic diagnostic study of *MIRAGE* behavior in VLM-based visual emotion recognition, identifying limitations beyond benchmark accuracy and offering new insights for future affective VLM evaluation and benchmark design. The code will be available at https://affective-mirage.github.io/Affective_Mirage/.

Recent vision-language models (VLMs), including both general-purpose [3–5] and affect-specialized models [6, 7], have demonstrated strong benchmark performance on visual emotion recognition. However, benchmark accuracy alone does not reveal how these models reach an affective perception [8], motivating a deeper investigation into their prediction behavior.

This concern is echoed by the recent *MIRAGE* phenomenon in VLMs [9], where models may produce seemingly competent predictions by implicitly imagining nonexistent visual content to answer the user query, rather than grounding their responses in the actual visual evidence. Existing studies [10–14] mainly examine conventional visual tasks in which relevant evidence is associated with identifiable objects or regions. Emotion recognition presents a more complex setting. Affective evidence can be distributed across facial expressions, body posture, nearby people, and the context of the scene [15–18], so affect at the scene-level can be incorrectly attributed to the target person [19]. Moreover, emotion labels are inherently subjective, and ambiguous expressions often produce substantial disagreement among human annotators [20].

We therefore investigate whether *MIRAGE*-like behavior also emerges in visual emotion recognition and how it differs from that observed in conventional visual tasks. As illustrated in Fig. 1, we propose *Affective MIRAGE* to diagnose this mechanism across three levels. **General mirage** asks what VLMs predict when visual evidence is unavailable in Diagnosis 1 (D1). **Causal mirage**¹ examines whether surrounding scenes substitute for removed person evidence in Diagnosis 2 (D2) and whether person or scene evidence governs predictions when the two conflict in Diagnosis 3 (D3). **Certainty mirage** investigates whether model uncertainty aligns with human disagreement on ambiguous emotions in Diagnosis 4 (D4). Together,

1. Introduction

Understanding human emotion from visual input is fundamental to socially aware artificial intelligence [1, 2].

¹We use *causal* in the intervention-based sense: changing subject or scene evidence while holding the other fixed and measuring its effect on model predictions. The scope is limited to model-level effects, not the causal generation of human emotion.

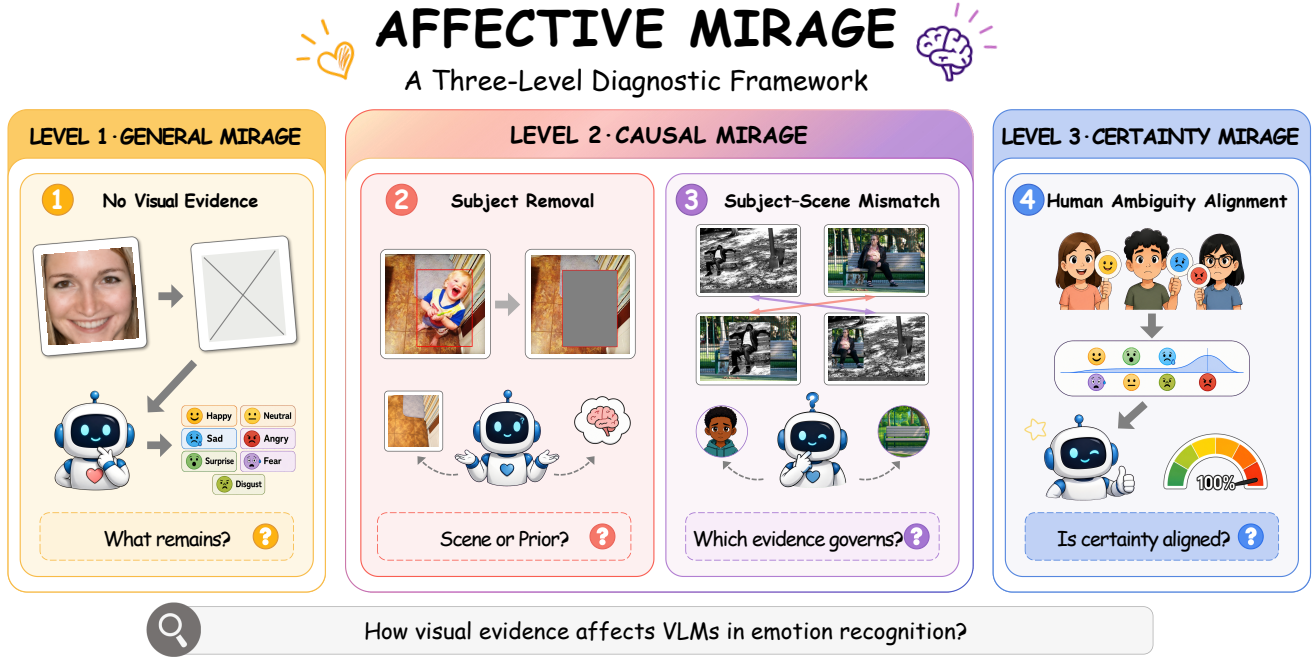


Figure 1. **Overview of our three-level diagnostic framework for affective mirage.** Level 1 examines predictions without visual evidence (D1), Level 2 probes causal reliance on subjects and scenes (D2 and D3), and Level 3 evaluates certainty alignment with human ambiguity (D4). To the best of our knowledge, this is the first work to systematically diagnose affective MIRAGE in VLM-based emotion recognition.

these diagnoses examine not only whether VLMs use visual evidence, but also which evidence they rely on and whether their certainty reflects the ambiguity of that evidence.

Across the three-level diagnostic framework, we conduct extensive experiments with seven state-of-the-art VLMs on four emotion recognition datasets. Three main findings emerge. First, MIRAGE-like behavior also exists in visual emotion recognition, but differs from conventional visual tasks: when visual evidence is unavailable, predictions exhibit clear **prediction collapse** in VLMs [21, 22], with distinct patterns toward the dominant emotion categories. Second, controlled removal and counterfactual recombination reveal a stronger causal influence of subject evidence than scene context on VLM predictions, supported by mechanism-level analyses. Third, model uncertainty partially tracks human affective ambiguity, but confidence remains systematically too high. Together, these findings reveal underexplored operating patterns of affective VLMs and provide implications for future model and benchmark design, as discussed in Sec. 5.

Our main contributions are summarized as follows:

- We propose a **three-level diagnostic framework** comprising general, causal, and certainty mirage to examine visual dependence, the causal influence of visual evidence, and uncertainty alignment. To our knowledge, this is the first systematic study of MIRAGE-like behavior in visual emotion recognition.

- Through controlled interventions and mechanism-level analyses, we uncover prediction collapse, the causal dominance of subject evidence over scene context, and partial uncertainty alignment with systematic overconfidence.
- We show that **benchmark accuracy alone is insufficient for evaluating affective VLMs**, translate these findings into guidance for affective VLM research, including no-vision controls, generality tests for specialized models, and multi-annotator benchmark design.

2. Related Works

2.1. Visual Emotion Recognition with VLMs

General VLMs enable visual emotion recognition through transferable image-text representations and instruction following [23–25]. Building on this paradigm, affective VLMs introduce emotion-specific multimodal adaptation [6, 7, 26], while recent benchmarks extend evaluation from emotion classification to affective understanding, reasoning, and generation [27–29]. Visual emotion can arise from both the target person and the surrounding context [30–32]. However, existing work primarily evaluates recognition accuracy or the plausibility of affective descriptions. In contrast, we diagnose VLM predictions by removing visual evidence, applying controlled interventions to subject and scene evidence, and testing whether model uncertainty reflects human affective ambiguity.

2.2. MIRAGE and Visual Grounding Diagnostics

Recent studies question whether strong VLM performance necessarily reflects genuine visual understanding. MIRAGE [9] shows that VLMs can generate image-grounded answers and retain nontrivial benchmark performance even when images are absent. Related work examines language priors through staged capability tests and counterfactual images [10], while visual counterfactuals have been used to separate reliance on image evidence from stored knowledge [33]. Mechanistic studies further probe where visual information enters the model and use internal interventions to examine its causal influence on generated outputs [34, 35]. These analyses mainly target general or medical visual reasoning. We extend this perspective to emotion recognition with diagnostics tailored to the distributed and subjective nature of affective evidence.

2.3. Ambiguity in Emotion Recognition

Emotion labels are inherently ambiguous, as the same facial expression may admit multiple human interpretations. This ambiguity is reflected in crowd-annotated datasets and emotion-distribution formulations that preserve disagreement beyond a single hard label [36–38]. Prior work addresses such ambiguity through distribution learning, label-correlation modeling, latent ambiguity mining, and uncertainty estimation [39–42]. [43] also studies ambiguous emotion recognition with LLMs in text. This work uses human disagreement as a diagnostic signal to test whether pre-trained VLM uncertainty aligns with human affective ambiguity beyond majority-label recognition accuracy.

3. Diagnostic Framework

3.1. Task Formulation

Emotion recognition with a VLM is defined as follows: given a sample i with visual modality V_i and input prompt T , the model predicts an emotion label $y \in \mathcal{Y}$,

$$y^* = \arg \max_{y \in \mathcal{Y}} p_{\mathcal{M}}(y | V_i, T), \quad (1)$$

where $p_{\mathcal{M}}(y | V_i, t)$ denotes the label-level prediction for class y under a specific inference scheme, and the VLM is parameterized by $\mathcal{M}$.

3.2. D1: General Mirage in Visual Absence

Diagnosis 1 examines how VLMs predict emotions when visual evidence is completely unavailable. For each facial image, we query the model under four stimulus conditions: *image_present*, which uses the original face; *blank*, which replaces it with a fixed 224×224 middle gray image; *noise*, which uses a Gaussian noise image deterministically generated from the sampling seed; *blank_mirage_hint*, which uses the blank image together with misleading metadata stating

that an image is attached. These conditions progressively remove visual evidence while testing whether either visual placeholders or an image presence cue can induce affective predictions, more details in Supplementary A (SM A).

Under conditions without visual evidence, ideally, we define **any valid emotion prediction as a mirage**, and an *abstention* represents an evidence-aware response from an ideal VLM, where the model explicitly recognizes that the visual information is unavailable. Some VLMs, especially specialized models, fail to parse or execute the task without visual input, which we define as a *parsing failure*. In this work, for simplicity, we consider both any valid emotion prediction and parsing failures as mirage responses as they both indicate a failure to appropriately handle the absence of visual evidence. The corresponding **Mirage Rate** R_{mirage} is the proportion of inference runs that still yield a valid emotion prediction or parsing failure.

To reliably assess the robustness of MIRAGE behavior to stochastic variation, in addition to a single-round verification, we also repeat each no-visual-evidence condition for S (50 in practice) independent runs, using a different random seed for each run. Each seed determines both the Gaussian noise realization, when applicable, and the sampling pattern (temperature=1.0).

As all evaluated VLMs exhibit prediction collapse under a fixed seed (Sec. 4.2), we further examine whether this behavior persists across random seeds. We define the collapse rate as:

$$R_{\text{collapse}} = \frac{1}{S} \sum_{s=1}^S \mathbb{I}[\hat{y}_s = y_{\text{collapse}}], \quad S = 50, \quad (2)$$

where y_{collapse} denotes the collapse label identified under the fixed-seed setting, which is *neutral* for most models.

The R_{mirage} measures whether the model fails to know the absence of visual evidence, whereas R_{collapse} measures the stochastic stability of predictions toward a predefined collapse label.

3.3. D2: Causal Mirage in Subject Removal

Diagnosis 2 tests whether VLM’s emotion prediction depends on visual evidence from the designated subject. Since visual emotion can be informed by both the subject and scene context [30–32], D2–D3 intervene on these two sources. For D2, we replace the subject bounding box b with a uniform gray region to obtain $\tilde{V}_i^b$ and compare

$$p_{\mathcal{M}}(y | V_i, T) \quad \text{and} \quad p_{\mathcal{M}}(y | \tilde{V}_i^b, T). \quad (3)$$

Systematic prediction changes reflect sensitivity to the subject-region evidence. Besides, Matched-mask and marker controls are provided in SM A and SM E. Intervention effects are assessed through recognition performance and prediction distributions, and additional analyses are provided in SM A and SM D.

Table 1. **The main results of D1–D4.** Acc@1 is reported for RAF-DB and CAER and Hit@1 for EMOTIC. For D1–D3, V_i is the original visual input, $\tilde{V}_i^b$ removes its subject, (S_i, C_j) combines the correct subject with a mismatched scene, and (S_j, C_i) the correct scene with a mismatched subject. R_{mirage} : MIRAGE rate. SS and MS denote single- and multi- seed settings, respectively. Parentheses show changes from V_i . For D4, ρ measures alignment between model and human entropy, while p_{top1} denotes mean prediction confidence.

Diagnosis	Dataset	Input / metric	GPT-5.6 Luna [44]	Gemini-3.5 Flash-Lite [45]	Qwen3.5 -9B [3]	InternVL3.5 -8B [4]	Ministral 3 8B [5]	Emotion -LLaMA* [6]	AffectGPT [7]
Diagnosis1	RAF-DB [37]	V_i	0.737	0.646	0.692	0.762	0.546	0.747	0.182
		No visual Acc	0.222 (−0.515)	0.222 (−0.424)	0.222 (−0.470)	0.222 (−0.540)	0.222 (−0.324)	0.222 (−0.525)	0.210 (−0.028)
		R_{mirage} (SS) [§]	100%	100%	100%	100%	100%	100%	100%
Diagnosis2 & Diagnosis3	EMOTIC [31]	R_{mirage} (MS) [§] (blank/noise/hint)	—	—	100%/100%/90%	100%/100%/100%	100%/97%/90%	100%/100%/100%	98%/100%/100%
		V_i	0.819	0.868	0.755	0.842	0.754	0.187	0.553
		$\tilde{V}_i^b$	0.437 (−0.381)	0.774 (−0.094)	0.487 (−0.269)	0.695 (−0.147)	0.563 (−0.191)	0.156 (−0.030)	0.550 (−0.003)
		(S_i, C_j)	0.670 (−0.148)	0.817 (−0.051)	0.493 (−0.262)	0.675 (−0.166)	0.513 (−0.240)	0.157 (−0.030)	0.559 (+0.006)
		(S_j, C_i)	0.225 (−0.593)	0.353 (−0.515)	0.246 (−0.509)	0.324 (−0.518)	0.385 (−0.369)	0.144 (−0.043)	0.549 (−0.004)
Diagnosis4 [†]	CAER [32]	V_i	0.528	0.540	0.530	0.416	0.456	0.220	0.347
		$\tilde{V}_i^b$	0.351 (−0.177)	0.351 (−0.189)	0.355 (−0.175)	0.346 (−0.069)	0.344 (−0.112)	0.127 (−0.093)	0.344 (−0.004)
		(S_i, C_j)	0.501 (−0.026)	0.532 (−0.007)	0.513 (−0.017)	0.412 (−0.004)	0.445 (−0.011)	0.219 (−0.001)	0.348 (+0.001)
		(S_j, C_i)	0.146 (−0.382)	0.191 (−0.349)	0.180 (−0.350)	0.104 (−0.311)	0.187 (−0.268)	0.107 (−0.113)	0.343 (−0.004)
Diagnosis4 [†]	FER+ [36]	ρ_{all}	0.403	0.357	0.601	0.548	0.233	0.365	−0.245
		ρ_{correct}	0.453	0.406	0.575	0.560	0.229	0.541	−0.077
		p_{top1}	0.681	0.734	0.855	0.843	0.822	0.547	0.992

[§] MIRAGE includes any emotion prediction or parsing failure; only abstention is non-MIRAGE. Table 2 reports the four component rates separately.

^{*} Emotion-LLaMA uses the EMOTIC target image or CAER’s first sampled frame, with unavailable auxiliary inputs zero-filled.

[†] For D4, closed-source results are reported for reference only because their output heads are inaccessible; comparisons are restricted to open-source models.

3.4. D3: Causal Mirage in Subject–Scene Mismatch

Because emotion may depend on multiple visual evidence sources, we perform an evidence-swapping diagnosis that independently swaps the subject region and scene context.

First, a sample i and a sample j are selected according to a pre-specified pairing rule (see SM A) such that their emotion annotations provide conflicting evidence. We conceptually decompose each visual input into the bounding-box subject region and the remaining scene context.

$$V_i = (S_i, C_i), \quad (4)$$

where S_i denotes the rectangular visual region inside the subject bounding box and C_i denotes the remaining visual content. Then, a crossed 2×2 source design is built:

$$\begin{array}{c|cc} & C_i & C_j \\ \hline S_i & (S_i, C_i) & (S_i, C_j) \\ S_j & (S_j, C_i) & (S_j, C_j) \end{array} \quad (5)$$

The diagonal combinations represent the original source pairings, whereas the off-diagonal combinations swap the subject and contextual sources between samples i and j .

To construct each crossed input, the original subject region is directly resized to the destination box using bicubic interpolation and pasted onto the target canvas, preserving the source region without cropping or padding. The same procedure is applied to both crossed conditions.

This crossed design further allows us to characterize which evidence dominates the resulting prediction. Let $\mathcal{E}$ denote the set of candidate visual evidence sources which may causally affect the prediction, and let $\mathcal{Y}_{e,i}$ denote the emotion annotations associated with the visual evidence source $e \in \mathcal{E}$ for sample i . Importantly, $Y_{e,i}$ can serve as an annotation proxy for source e , rather than necessarily constituting an independent ground-truth annotation of that source. Considering that the predicted label y_i may fall into the annotation sets of multiple visual evidence sources, we assign y_i to the source e when it belongs only to $\mathcal{Y}_{e,i}$ and not to any other annotation sets:

$$a_{i,e} = \mathbb{I} \left[y_i^* \in \mathcal{Y}_{e,i} \setminus \bigcup_{k \in \mathcal{E}, k \neq e} \mathcal{Y}_{k,i} \right]. \quad (6)$$

Then, we define the **Causal Evidence Dominance (CED) Score** of e as

$$D_e = \frac{1}{N} \sum_{i=1}^N a_{i,e} - \max_{k \in \mathcal{E}, k \neq e} \frac{1}{N} \sum_{i=1}^N a_{i,k}, \quad (7)$$

where the first term indicates the number of samples aligning with e , and thus the second term represents its strongest competing visual evidence source. Accordingly, a positive D_e indicates that e dominates all competing visual evidence sources in directing the model prediction, with larger values

indicating stronger dominance over its strongest competitor.
The dominant visual evidence source is therefore

$$e^* = \arg \max_{e \in \mathcal{E}} \frac{1}{N} \sum_{i=1}^N a_{i,e}. \quad (8)$$

Predictions that cannot be uniquely attributed to any
source are treated as unresolved, with rate

$$F_U = 1 - \frac{1}{N} \sum_{i=1}^N \sum_{e \in \mathcal{E}} a_{i,e}. \quad (9)$$

In our subject-context intervention, $\mathcal{E} = \{S, C\}$, where
S and C denote the subject and scene-context evidence
sources, respectively. In this context, for the crossed in-
put (S_j, C_i) , $\mathcal{Y}_{s,i}$ uses the subject annotation of sample j ,
while $\mathcal{Y}_{c,i}$ uses the subject annotation of sample i itself, as
no independent scene-only annotation is available. Here,
 $Y_{c,i}$ is a source-associated proxy rather than a scene-only
ground truth; thus, CED measures relative subject–context
alignment under counterfactual recombination, not intrinsic
scene affect. Since our setting contains only two evidence
sources, $D_C = -D_S$. We therefore report only D_S .

Similar to Sec. 3.3, we additionally conduct a marker-
free subject-region replacement control to disentangle the
effect of visual manipulation from explicit attention guid-
ance. See SM A for details.

3.5. D4: Certainty Mirage in Affective Ambiguity

The final diagnosis examines whether VLMs exhibit affec-
tive uncertainty that reflects ambiguity in human emotion
judgments.

For open-source models, we use teacher forcing to ob-
tain the log-likelihood $\ell_{i,c}$ of each candidate emotion label
 y_c given the image and prompt x_i . These scores are normal-
ized over the candidate emotion space to form the model
class distribution q_i :

$$\ell_{i,c} = \log p_\theta(y_c | x_i), \quad q_{i,c} = \frac{\exp(\ell_{i,c})}{\sum_{c' \in \mathcal{Y}} \exp(\ell_{i,c'})},$$

$$H_M(i) = - \sum_{c \in \mathcal{Y}} q_{i,c} \log_2 q_{i,c}. \quad (10)$$

Here, $q_{i,c}$ denotes the normalized model probability as-
signed to class c . $H_M(i)$ is the entropy of the model pre-
diction, with higher values indicating greater affective un-
certainty. We additionally report $p_{\text{top1}}(i)$, the probability
assigned to the predicted class, as a measure of prediction
confidence. Full details of the candidate scoring procedure
are provided in the SM A.

Closed-source models are analyzed separately using
their self-reported structured confidence distributions, since
they do not expose hidden states or vocabulary logits. These

results are reported for reference only and are not used to
support our main conclusions.

To obtain the human reference, we normalize the 10-
annotator votes for each image into a distribution $\mathbf{h}_i$ and
compute its entropy $H_H(i) = - \sum_c h_{i,c} \log_2 h_{i,c}$. A larger
 $H_H(i)$ represents greater disagreement among human an-
notators. We measure human-aligned affective uncertainty
using the Spearman correlation $\rho_S(H_H, H_M)$, which tests
whether the model becomes more uncertain on images that
humans also perceive as emotionally ambiguous.

4. Experiments

4.1. Experimental Setup

Each diagnosis requires different visual evidence and anno-
tations, so we use the benchmark best suited to each objec-
tive and use its validation/test sets. Although each diagnosis
uses a single dataset or designed datasets, the observed be-
haviors remain informative for the mirage examined.

For D1, we use RAF-DB [37] (7-class, single-label)
and report Acc@1 under *image-present*, *blank*, *noise*, and
mirage-hint conditions. For D2–D3, we use EMOTIC
[31] (26-label, multi-label; Hit@1) and CAER [32] (7-
class, single-label; Acc@1). As CAER lacks subject boxes,
we obtain pseudo subject regions with YOLOv8 [46];
face/head-region controls using [47, 48] are in SM G. For
D4, we use FER+ [36], whose ~ 10 crowd annotations per
image offer human label distributions, and report Spearman
 ρ between model and human entropy, with majority-label
accuracy as a prerequisite. More details are in SM G.

We evaluate three general VLMs, Qwen3.5-9B [3],
InternVL3.5-8B [4], and Ministral-3-8B [5]; two affec-
tive VLMs, AffectGPT [7] and Emotion-LLaMA [6], as
domain-specific representatives; and two closed-source
models, GPT-5.6-Luna [44] and Gemini-3.5-Flash-Lite
[45]. The input prompt remains consistent within the
dataset. Experiments are run on AMD MI250X (D2/D3)
and NVIDIA GH200 (D1/D4) GPUs.

4.2. General Mirage: Label Prior Collapse

As shown in Table 1 and Table 2, most models perform well
when visual evidence is available, with five of the seven ex-
ceeding 0.64 Acc@1. Once visual input is removed, most
models collapse toward predicting *neutral*, reaching an ac-
curacy of 0.222 (the proportion of neutral samples in RAF-
DB). **Crucially, the models rarely acknowledge the ab-
sence of visual evidence and instead fall back to a domi-
nant label prior.** As we can see, every open-source model
exhibits a **near-100% mirage rate**, even across 50 seeds
and the three no-vision configurations; abstentions remain
vanishingly rare, keeping the mirage rate at 90–100%. In
other words, VLMs almost never signal that the image is
absent, and they instead assert a *neutral* prior. Unlike the

Table 2. **D1: Single-seed/multi-seed output rates without vision.** Each entry reports the seed-0 rate / mean rate over $S = 50$ seeds. The four mutually exclusive rates sum to 1: *Col.to.neu*, *neutral* output; *Off-neu*, non-neutral output; *P-fail*, parsing failure; and *Abst*, abstention. Only *Abst* is non-MIRAGE.

Model	Cond.	Valid emo. output		P-fail	Abst	R_{collapse}	R_{mirage}
		Col.to.neu	Off-neu				
InternVL3.5-8B	Blank	1/1	0/0	0/0	0/0	1/1	1/1
	Noise	1/1	0/0	0/0	0/0	1/1	1/1
	Hint	1/1	0/0	0/0	0/0	1/1	1/1
Qwen3.5-9B	Blank	1/1	0/0	0/0	0/0	1/1	1/1
	Noise	1/1	0/0	0/0	0/0	1/1	1/1
	Hint	1/0.90	0/0	0/0	0/0.10	1/0.90	1/0.90
Ministral-3-8B	Blank	1/0.90	0/0	0/0.10	0/0	1/0.90	1/1
	Noise	1/0.97	0/0	0/0	0/0.03	1/0.97	1/0.97
	Hint	1/0.90	0/0	0/0	0/0.10	1/0.90	1/0.90
Emotion-LLaMA	Blank	1/0.70	0/0.30	0/0	0/0	1/0.70	1/1
	Noise	1/0.50	0/0.50	0/0	0/0	1/0.50	1/1
	Hint	1/0.70	0/0.30	0/0	0/0	1/0.70	1/1
AffectGPT	Blank	0.94/0.44	0/0	0.06/0.54	0/0.02	0.94/0.44	1/0.98
	Noise	1/0.54	0/0	0/0.46	0/0	1/0.54	1/1
	Hint	1/0.30	0/0.02	0/0.68	0/0	1/0.30	1/1

Table 3. **D3: CED under subject replacement** (S_j, C_i). Positive D_S indicates subject-evidence dominance; F_U : unresolved rate.

Model	EMOTIC		CAER	
	D_S	F_U	D_S	F_U
GPT-5.6 Luna	+32.4%	22.5%	+27.4%	43.4%
Gemini-3.5 Flash-Lite	+19.4%	10.0%	+19.7%	42.2%
Qwen3.5-9B	+27.8%	23.0%	+22.1%	41.8%
InternVL3.5-8B	+16.9%	18.3%	+35.6%	43.5%
Ministral 3 8B	-3.0%	26.0%	+19.4%	43.1%
Emotion-LLaMA	+6.2%	65.1%	+29.9%	48.7%
AffectGPT	-13.3%	3.5%	-22.1%	53.4%

unconstrained guessing reported in conventional MIRAGE settings [9], label-constrained emotion recognition exhibits a distinct failure mode: a concentrated prediction collapse to the dominant label. We also observe that domain-specific VLMs, such as AffectGPT, contain a different mirage issue (parsing failure). It dips to ~ 0.210 because a small fraction of its outputs violate the required format rather than collapsing to *neutral* label; across seeds, AffectGPT frequently produces parsing failures under some conditions. We treat this as an AffectGPT-specific fitting issue that does not affect the overall trend (SM B).

Table 2 also reports R_{collapse} , the rate of predictions falling into the identified dominant label, *neutral*. Notably, general VLMs largely collapse cleanly to *neutral* across seeds, with both collapse and MIRAGE rates close to 1. In contrast, emotion-specific models retain MIRAGE rates at or near 1 but exhibit substantially lower collapse rates. The causes differ: Emotion-LLaMA predicts other emotion labels, whereas AffectGPT frequently produces parsing failures. We further discuss these distinct behaviors in SM C.

Key Finding 1. Without visual evidence, VLMs maintain a nearly 100% mirage rate in emotion recognition, while their predictions collapse toward a dominant label. Emotion specific models introduce greater variation in this collapse pattern.

4.3. Causal Mirage I: Prediction Collapse

Diagnosis 2 examines model behavior when the subject evidence is removed. As shown in Table 1, subject removal reduces performance for the five general VLMs by 0.094-0.381 on EMOTIC and 0.069-0.189 on CAER, whereas AffectGPT changes little and Emotion-LLaMA shows only small declines, suggesting different patterns. Furthermore, Macro-F1 degrades more strongly (see SM D).

VLM predictions exhibit clear prediction collapse. Fig. 2 shows that prediction distributions are already skewed under the original input. For instance, despite EMOTIC having 26 labels, the three most frequently predicted labels account for most of GPT-5.6 Luna’s predictions. Subject removal further amplifies this pattern, driving predictions toward a smaller set of dominant labels, especially on CAER. Detailed statistics in SM D further show where this collapse occurs: the five general VLMs predominantly shift toward *Disconnection* or *Engagement* on EMOTIC, while all collapse toward *Neutral* on CAER. The CAER face/head control confirms that this is not solely due to large subject regions (SM E). In contrast, Emotion-LLaMA exhibits a distinct pattern from the other VLMs: it collapses to *Affection* on EMOTIC and to *Surprise/Anger* on CAER. Above all, Diagnosis 2 reveals the affective causal mirage: non-trivial Hit@1 or Acc@1 can persist after removing subject evidence because the recognition score is partly sustained by prediction collapse rather than effective substitution by scene evidence.

Controlled experiments further support this interpretation (see SM E). Moving the same gray box away from the subject largely restores performance, ruling out generic occlusion effects. Subject removal also has a stronger effect in single-person images and weakens as more subjects are visible, consistent with compensation from remaining person-level evidence.

Key Finding 2. VLMs exhibit obvious prediction collapse on emotion prediction tasks, which can be amplified by removing subject-region evidence, though they still achieve nontrivial scores on benchmarks.

4.4. Causal Mirage II: Subject Evidence Dominance

Diagnosis 3 compares the causal influence of subject and scene evidence using the two crossed conditions (S_i, C_j) and (S_j, C_i). Table 1 shows larger performance drops under subject replacement than scene context replacement: 0.369-0.593 vs. 0.051-0.262 on EMOTIC, and 0.268-0.382 vs. 0.004-0.026 on CAER. See detailed analysis in SM D.

Table 3 further identifies the dominance of the visual

evidence sources. CED score is positive for four of five general VLMs on EMOTIC and all five on CAER, indicating that when subject and scene evidence conflict, general VLMs usually follow the subject region. In contrast, two affective VLMs are less consistent: Emotion-LLaMA has a positive CAER score but a high unresolved rate, while AffectGPT follows the retained scene context more often than the inserted subject region on both datasets. Together with Diagnosis 2, these results suggest that scene context rarely substitutes for person-specific affective evidence.

Besides, control analyses confirm that the subject dominance is not caused by large pseudo-person boxes or explicit target marking (see SM E). On CAER, even replacing only the localized face/head region affects predictions substantially more than replacing the remaining context across all five general VLMs. On single-person EMOTIC samples, removing the marker has only a minor effect compared with replacing the subject-region content, indicating that the observed preference is primarily driven by visual evidence within the subject region. These controls therefore support the subject-region dominance observed for general VLMs. See SM E for complete control results for all models.

Furthermore, we examine whether the strong behavioral dependence on subject evidence is reflected in the model’s internal states on InternVL3.5-8B, Qwen3.5-9B, and Emotion-LLaMA. We patch visual-token states from the full-input inference into the subject-masked inference and test whether they restore the full-input prediction. For InternVL3.5-8B and Qwen3.5-9B, restoring subject-region-related states recovers the original-input prediction on substantially more samples than restoring states at the remaining image positions (See SM F). Fig. 3 visualizes this effect through prediction restoration maps (PRMs) at 25% decoder depth, where higher values indicate a larger increase in the probability of the full-input prediction after restoring that visual-token state. Thus, the sensitivity to subject evidence is accompanied by a localized internal dependence on subject-associated visual states, providing mechanism-level support for our subject-grounding finding. The algorithm and detailed results are provided in SM F.

Key Finding 3. General VLMs rely predominantly on subject evidence, with scene context failing to substitute when the subject is removed. Affective VLMs show more model-specific source preferences, indicating that emotion specialization does not ensure more consistent causal grounding.

4.5. Certainty Mirage: Limited Uncertainty Alignment

Diagnosis 4 examines whether model uncertainty increases with human disagreement in emotion judgments. As shown in Table 1, four of the five open-source models exhibit significantly positive correlations between model and hu-

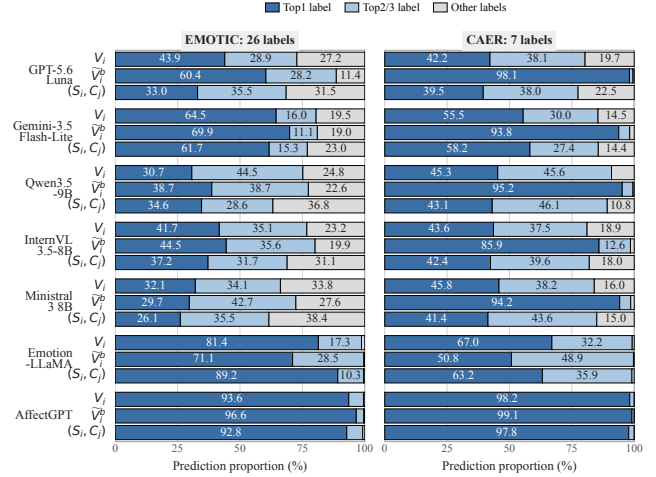


Figure 2. D2&D3: Prediction concentration under the original input, subject removal, and context replacement. Top1 label means the most frequently predicted label; Top2/3 labels denote the second- and third-most frequently predicted labels. Greater concentration in Top1/3 indicates stronger prediction collapse.

man entropy. Qwen3.5-9B and InternVL3.5-8B achieve the highest ρ_{all} values of 0.601 and 0.548, respectively, indicating partial alignment with human affective ambiguity. AffectGPT instead produces a negative correlation of -0.245 , consistent with its behavior in D1.

A positive ρ_{all} could arise simply because models are confident when correct and uncertain when incorrect. However, ρ_{correct} remains positive for the same four models, while AffectGPT remains close to zero. Emotion-LLaMA even increases from 0.365 to 0.541 on correctly predicted samples. Partial correlations controlling for prediction correctness are also significantly positive, showing that the observed alignment cannot be explained solely by correctness.

Rank alignment, however, does not imply matched confidence levels. We divide samples into high human agreement ($h_{\text{human}} \geq 0.8$) and low human agreement ($h_{\text{human}} \leq 0.5$) groups. The high agreement threshold corresponds to the median of 0.800, while the low agreement threshold identifies samples without a clear majority. As shown in Table 4, the accuracy of the four positively correlated models drops substantially in the low agreement group, whereas p_{top1} decreases much less and remains higher than accuracy. Emotion-LLaMA has the lowest overall p_{top1} of 0.547, while AffectGPT becomes more confident on the low agreement group.

Figure 4 further compares model and human entropy. For the three general-purpose models, many samples with high human entropy fall below the diagonal, indicating lower model uncertainty when human judgments are highly ambiguous. Emotion-LLaMA exhibits broadly higher entropy, including on samples with low human entropy, while

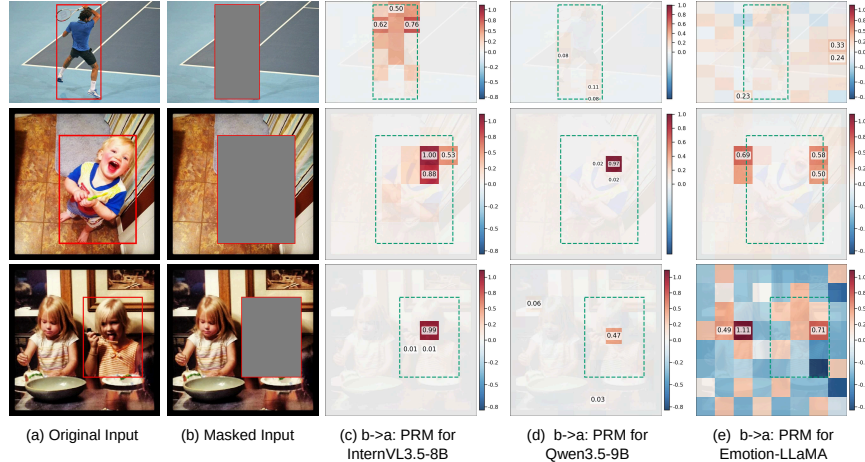


Figure 3. **Case Study: PRMs for three EMOTIC samples at the 25% depth of each VLM’s decoder.** Each row shows the original and subject-masked inputs followed by PRMs from three VLMs. The three rightmost columns show PRMs from different VLMs. The manually drawn green dashed rectangle marks the subject region. Higher PRM values indicate a larger increase in the probability of the full-input prediction after restoring that visual-token state. Maps are model-specific due to different visual token grids.

Table 4. **D4: Accuracy and confidence across human agreement groups.** H and L denote high ($h_{\text{human}} \geq 0.8$) and low ($h_{\text{human}} \leq 0.5$) agreement, respectively.

Model	p_{top1}	Acc. H	Acc. L	p_{top1} H	p_{top1} L
Qwen3.5-9B	0.855	0.936	0.472	0.923	0.723
InternVL3.5-8B	0.843	0.877	0.371	0.905	0.739
Ministral-3-8B	0.822	0.743	0.356	0.855	0.786
Emotion-LLaMA	0.547	0.608	0.238	0.620	0.475
AffectGPT	0.992	0.251	0.364	0.875	0.913

AffectGPT remains concentrated at low model entropy. Overall, the models capture a coarse ranking of sample difficulty without fully matching the level of human affective ambiguity. Further discussion is provided in SM B.

Key Finding 4. Model uncertainty generally increases with human disagreement, but confidence remains systematically too high, indicating only partial alignment with human affective ambiguity.

5. Discussion and Conclusion

We present the first systematic diagnosis of MIRAGE-like behavior in visual emotion recognition, revealing label-prior collapse without vision, person-dominant grounding under intervention, and partial but overconfident uncertainty alignment.

Future work can extend the diagnosis to additional evidence sources such as background, illumination, and other contextual cues. Besides, while our prompts may reinforce a visual-answering regime, consistent collapse across blank, noise, and blank mirage hint inputs suggests prompting may

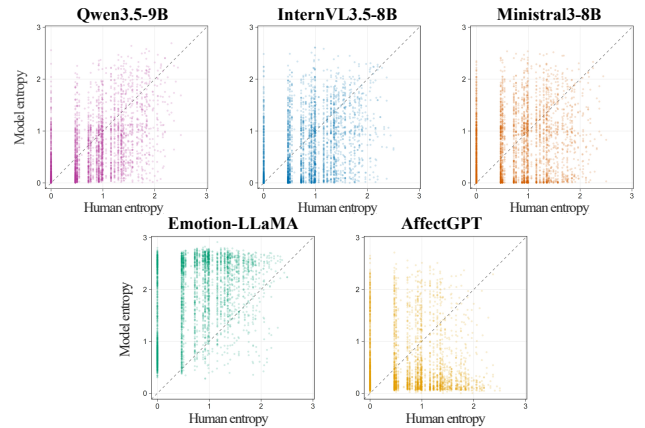


Figure 4. **D4: Comparison of model and human entropy.** Each point represents one sample. Emotion-LLaMA exhibits generally higher uncertainty, whereas the other models show lower entropy.

amplify MIRAGE [9] but is unlikely to be its sole cause, which can be further explored.

Our findings suggest three practical recommendations. First, affective VLMs should be evaluated under missing or uninformative visual inputs, with prediction distributions reported alongside accuracy. Second, emotion-specialized models should be tested across datasets, label spaces, and evidence conditions to assess whether specialization improves grounding rather than merely reinforcing label priors. Third, benchmarks should explicitly incorporate human ambiguity and uncertainty. Together, these diagnoses expose limitations obscured by benchmark accuracy and provide guidance toward more grounded and reliable affective VLMs.

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